

CHE 205, Fall 2019, Problem Set 8

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Due date: 11:59pm, Wed, 11/20/2019

All problems are from the textbook by V.L. Law.

Submit two spreadsheets, one for each problem, on blackboard.

Problem 1 (10 pts). Exercise 8.1b using forward Euler method in time, and central difference method in space. Set $0 \leq x \leq 1$, $\Delta x = 0.05$, $\Delta t = 0.001$, and $t_{max} = 0.3$. Plot $u(x, t = 0)$ (blue), $u(x, t = 0.1)$ (red), $u(x, t = 0.2)$ (green), and $u(x, t = 0.3)$ (purple) against x .

Problem 2 (10 pts). Repeat problem 1 above, now this time using the Crank-Nicholson method (hint: use and modify the VBA script of example 8.4 in the text).

Example 8.1: Transient Heat Conduction in a Rod

Shown below are the first few rows of an Excel program to solve Equation 8.10 together with the initial and boundary conditions given in Equation 8.7:

[illegible]

Values of $\Delta\theta$ of 0.01, 0.001, and 0.0001 were tried. $\Delta\theta = 0.001$ gave nearly the same results as $\Delta\theta = 0.0001$. The value of u corresponding to $\theta = 0.1$ and $x = 0.5$ was 0.7361. This compares favorably with the analytically computed value of 0. 0.7372. When $\Delta\theta = 0.001$, the numerically computed value was 0.7371. Given the simplicity of this method, these results are remarkably good. It must be pointed out, however, that when $\Delta\theta = 0.01$, the solution was unstable (wildly fluctuating temperatures were the result).

Exercise 8.1: Rework the problem of Example 8.1 using the simple explicit method and the following initial and boundary conditions:

- a. $u(x, 0) = 1; u(0, t) = u(1, t) = 0$
- b. $u(x, 0) = 1; u(0, t) = 0; du(1, t)/dx = 0$ (right side insulated)
- c. $u(x, 0) = x$ (linear initial temperature profile); $u(0, t) = 1; u(1, t) = 0$
- d. $u(x, 0) = \sin(\pi x); u(0, t) = u(1, t) = 0$
- e. $u(x, 0) = x; du(0, t)/dx = du(1, t)/dx = 0$